THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET C

Chapters 9–12: **Light – Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. A ray of light passing through the optical centre of a lens: 1

(a) goes undeviated **(b)** passes through the focus
(c) becomes parallel to the axis **(d)** reflects back

2. The SI unit of power of a lens is: 1

(a) dioptre **(b)** joule **(c)** watt **(d)** metre

3. The focal length of a plane mirror is: 1

(a) 1 m **(b)** infinite **(c)** zero **(d)** negative

4. The splitting of white light into its component colours is called: 1

(a) refraction **(b)** dispersion **(c)** scattering **(d)** reflection

5. The defect corrected by bifocal lenses is: 1

(a) myopia only **(b)** hypermetropia only
(c) presbyopia **(d)** cataract

6. The phenomenon responsible for the blue colour of the sky is: 1

(a) reflection **(b)** scattering of light
(c) dispersion **(d)** refraction

7. The heating effect of current is given by: 1

(a) $H = V/R$ **(b)** $H = IRt^2$ **(c)** $H = IR^2t$ **(d)** $H = I^2Rt$

8. The material used for the filament of an electric bulb is: 1

(a) nichrome **(b)** copper **(c)** tungsten **(d)** aluminium

9. The slope of a V–I graph gives: 1

(a) power **(b)** resistance **(c)** current **(d)** charge

10. Inside a bar magnet, field lines go from: 1

(a) north to south **(b)** south to north
(c) east to west **(d)** they do not exist

11. In Fleming's left-hand rule, the middle finger indicates the direction of: 1

(a) current **(b)** magnetic field **(c)** motion **(d)** force

12. Which of these is used to make a strong electromagnet? 1

(a) soft iron core **(b)** copper core **(c)** wooden core **(d)** steel core

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Why does a ray of light bend when it travels from one medium to another? 2

14. What is the far point and near point of a normal human eye? 2

15. Why are alloys used in heating devices? 2

16. A student says the field at the centre of a circular loop is zero. Justify or refute. 2

17. Name two safety devices used in domestic circuits. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. An object is placed 20 cm in front of a concave mirror of focal length 15 cm. Find the position and nature of the image and the magnification. 3

19. Explain the formation of a rainbow with a diagram (describe). 3

20. Draw (describe) a circuit diagram with a cell, a resistor, an ammeter, a voltmeter and a plug key. How are the ammeter and voltmeter connected? 3

21. State Fleming's left-hand rule. Name two factors on which the force on a conductor depends. 3

22. What is a rainbow? How is it formed? 3

OR

Explain why the Sun appears reddish at sunrise and sunset. 3

23. What is resistivity? On what factors does the resistance of a conductor depend? 3

OR

Find the heat produced in 10 s by a current of 2 A through a resistor of 50 Ω. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) Explain atmospheric refraction with two examples. 5
(ii) Why does the apparent position of a star change?
(iii) Why do the objects seen through hot air appear to waver?

OR

(i) What is a spectrum? 5
(ii) Why does white light split in a prism?
(iii) Draw (describe) the arrangement to recombine the spectrum.
(iv) Name the colour with maximum deviation.

25. (i) Draw (describe) the schematic diagram of a domestic electric circuit. 5
(ii) Why are two separate circuits used (15 A and 5 A)?
(iii) Role of the earth wire and fuse.

OR

(i) Explain the magnetic field due to a solenoid with a diagram. 5
(ii) How can a solenoid be used to magnetise a steel rod?
(iii) Difference between electromagnet and permanent magnet.

END OF QUESTION PAPER — Total: 50 Marks